local IP (172.16.20.101).

```
In short, [Elastic IP]:522 π 172.16.20.101:522  π
192.168.20.5:22
```

This means that you can SSH the OpenStack VM globally by using the elastic IP address and the respective port number.

Elastic IP address: For this type of customised OpenStack installation, we require at least two NICs for an AWS EC2 instance. One is for accessing the VM terminal for installing and accessing the dashboard. In short, it acts as a management network, a VM tunnel network or an API network. The last one is for an external network with a unique type of interface configuration and is mapped with the provider network bridge (say *br-ex* with *eth1*).

AWS will not allow any packets to travel out of the VPC unless the elastic IP is attached to that IP address. To overcome this problem, we must attach the elastic IP for this NIC. This is so that the packets of the OpenStack's VM reach the OpenStack router's gateway and from the gateway, the packets get embedded with the registered MAC address. Then, the matching IP address will reach the AWS switch (VPC environment) via *br-ex* and *eth1* (a special type of interface configuration), and then hit the AWS actual VPC gateway. From there, the packets will reach the Internet.

Other cloud platforms

In my analysis, I noticed that most of the cloud providers like Dreamhost and Auro-cloud have the same limitations for OpenStack networking. So we could use the tricks/hacks mentioned above in any of those cloud providers to run an OpenStack cloud on top of them.

Note: Since we are using the QEMU emulator without KVM for the nested hypervisor environment, the VM's performance will be slow.

If you want to try OpenStack on AWS, you can register for CloudEnabler's Cloud Lab-as-a-Service offering available at *https://claas.corestack.io/* which provides a consistent and on-demand lab environment for OpenStack in AWS. END

References

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